Creating a Multilingual Dataset in Arabic and Croatian from Sports Videos Through a Data Processing Pipeline Combining ASR and MT

Wajdi Zaghouani[1](✉), Sanja Seljan[2], Ivan Dunđer[2], Rashid Yahiaoui[1], and Amer Al-Adwan[1]

[1] Northwestern University in Qatar, Education City, Doha, Qatar
wajdi.zaghouani@northwestern.edu, {ryahiaoui,aadwan}@hbku.edu.qa
[2] Faculty of Humanities and Social Sciences, University of Zagreb, Ivana Lučića 3, 10000 Zagreb, Croatia
{sanja.seljan,idundjer}@ffzg.unizg.hr

Abstract. Machine translation is getting more and more attention in the research community that deals with natural language processing, language resources and language technologies. It is considered to be one of the most important disruptive technologies with immense implications and benefits for mankind. Closely related is the field of speech technologies that enable tasks, such as automatic speech recognition and speech generation. Both machine translation and automatic speech recognition are explored in this research. The main goal of this paper is to examine the possibilities and obstacles of combining automatic speech recognition with machine translation in a web-based audio-video environment, and in a real-time setting in the sports domain that covers football matches for the purpose of creating a multilingual dataset. The research is performed for two language pairs, English-Arabic and English-Croatian. Captions from videos that contain live sports comments were automatically generated by an automatic speech recognition approach, then machine-translated by a popular online machine translation service, and afterwards edited in three distinct processing phases that considered different aspects of human involvement. Quality evaluations are performed by native speakers with regard to the criterion of usability, and by applying BLEU, the most prominent automatic machine translation quality metric today.

Keywords: Machine Translation · Post-Editing · Automatic Speech Recognition · Terminology · Quality Evaluation · Multilingual Dataset Creation · Human Evaluation · BLEU · Data Processing Pipeline

1 Introduction

The amount of audio-video content is growing rapidly, as well as the demand to access multimedia resources in various languages, or to disseminate multimedia resources to a wide public. The volume of data to be retrieved and transferred in various languages, language pairs and specific domains represents a considerable challenge for innovative technologies due to the great linguistic complexities.

© The Author(s), under exclusive license to Springer Nature Switzerland AG 2025
B. Hdioud and S. L. Aouragh (Eds.): ICALP 2024, CCIS 2340, pp. 182–195, 2025.
https://doi.org/10.1007/978-3-031-80438-0_14

Machine translation (MT) is increasingly gaining more attention in the research community that deals with natural language processing, language resources, language technologies, artificial intelligence, and machine learning. It can be understood as a complex task in which a computer is utilized for the purpose of translating from the source to one or more target languages without human involvement, or with a minimum of interventions [6]. High-quality machine translation systems are usually designed for a specific language pair and domain, such as for industry [5], poetry [6], philosophy and religion [19], etc. Quality evaluation is a vital aspect of machine translation assessment [6, 7]. Closely related is the study of speech technologies that enable automatic speech recognition (ASR) and speech generation [18].

ASR systems transcribe speech to text using acoustic and language models as in [16]. Recent neural network architectures have driven improvements, but accuracy still drops for informal, conversational speech [17]. Machine translation approaches like statistical and neural models translate text across languages [13]. However, they rely on wellformed input and often struggle with domain-specific terms. Live sports commentary combines challenges from both tasks – excited speech with domain terminology – but has potential applications for expanding media access globally [3, 8].

This research explores the English-Arabic and English-Croatian language pairs to cover linguistically distant languages. The morphologically rich structure of Arabic and Croatian can worsen ASR and MT performance [11].

Using a real-world football dataset allows evaluating the robustness of the combined system. Both machine translation (MT) and automatic speech recognition (ASR) are therefore examined in this research. Quality evaluation in this paper is performed on raw machine translations, then on post-edited machine translations and, finally, on segmented machine translations in order to assess the usability and effectiveness of machine translation technology in a real-time setting that covers a football world cup event organized by FIFA. The main goal of this paper was to examine the possibilities and obstacles of combining automatic speech recognition with machine translation in a web-based audio-video environment, and in a real-time setting in the sports domain that covers football matches for the purpose of creating a multilingual dataset.

The main contributions of this paper are as follows: (1) Creation of a multilingualin-domain dataset by using a data processing pipeline which combines ASR and MT; (2) Analysis of the data pipeline on live sports commentary speech, which poses challenges for both technologies; (3) Experimentation with multi-phase processing incorporating text corrections and logical segmentation to improve downstream MT quality; (4) Evaluation of the approach on English-Arabic and English-Croatian using automatic metrics and human judgments of usability. The results provide insights into the current capabilities and limitations of using ASR and MT for real-time multilingual speech translation in a specific domain of application.

The paper is structured in the following way. In the first section the authors highlight the impact of machine translation and speech technologies, whereas the second section presents relevant related work. The research methodology is described in the third chapter, with special emphasis on evaluation of automatic speech recognition, followed by evaluation of raw, post-edited and segmented machine translations. Research results are discussed in the fourth chapter. The fifth chapter considers the limitations of

this paper. In the sixth chapter conclusions and suggestions for future research are stated. Finally, the seventh and eighth chapter provide more information on data availability and ethics.

2 Related Work

The foundational insights into evaluation campaigns for machine translation provided by Kraif et al. [24] offer a vital historical and methodological perspective that is fundamental for any study in this domain. This background is particularly relevant given the challenges and intricacies involved in accurately evaluating MT systems.

Alam et al. [25] explore check-worthiness in multimodal content, providing a nuanced understanding of integrating various language technologies. This work is crucial for your research, especially when considering the combination of speech recognition and translation. Atwell et al. [26] discuss leveraging computational methods to understand complex texts, particularly in Arabic. While their focus is on religious texts, the computational approaches they discuss are applicable to the processing and understanding of Arabic in diverse contexts, including sports commentary. Draffan et al. [27] provide insights into speech technology applications for AAC users, offering a unique perspective that could enrich the speech recognition discussion in your study, particularly in the context of real-time applications. Schlippe et al. [28] investigate visualizing voice characteristics, offering valuable perspectives on processing speech data, essential for the ASR component of your research.

Moreover, the work by Seljan & Zaghouani [29] on integrating MT technology into AV web-based environments aligns closely with your study's objectives, offering direct insights into the practical aspects of combining ASR and MT in live settings. Additionally, Zaghouani et al.'s [30] contribution to building an Arabic Machine Translation Post-Edited Corpus underscores the importance of tailored annotation guidelines in enhancing MT quality, particularly for Arabic, a language of focus in your study. Temnikova et al.'s [31] exploration of cognitive evaluation approaches in MT, and their further development by Zaghouani & Temnikova [32], provide critical perspectives on MT quality assessment, particularly relevant for your study's emphasis on usability and quality evaluations. Lastly, Zaghouani et al.'s [33] work on normalizing mathematical expressions to improve translation underscores the significance of domain-specific adaptations in MT, analogous to the sports terminology challenges in your research.

Existing research related to integration of machine translation into audio-video technology is scarce, and does not cover the domain of sports or real-time solutions for dissemination of multilingual information for numerous low-resourced languages.

Armstrong, Caffrey, and Flanagan [2] performed research on the translation of subtitles for movies released on DVDs, focusing on the use of example-based machine translation, which can be edited afterward. Evaluation results show domain-related problems with the training corpus, problems with chunking of subtitles, and recognition of nonverbal utterances.

Martín-Mor and Sánchez-Gijón [15] discussed the integration of machine translation into localization and audio-visual products, particularly in voiceover documentaries, which could boost the spreading of non-commercial products, and improve the dissemination of cultural information. Results show better scores for texts that are more formal,

having a distinct direct style with previously recorded sentences and without ambient sound. Automatic quality evaluation with metrics, such as BLEU, TER, METEOR, and F-measure show similar results, however giving higher scores to narration when compared to dialogues.

Seljan and Katalinić [20] analyzed possible automation methods for localization of audio-video material, pointing out the need for quality assurance. Burchardt et al. [3] analyzed potentially useful methods and tools for measuring machine translation quality in a fully integrated real-time web-based translation environment.

Etchegoyhen et al. [8] presented a large-scale quality evaluation of machine translation for professional subtitling and a productivity gain-loss analysis. The main research drawback was the cognitive effort in assessing low-quality machine translation output. Volk [23] analyzed text genre characteristics of film subtitles, emphasizing quality needs and customer satisfaction.

An analysis of a process that combined automatic speech recognition and machine translation, which required specific-domain training, is presented in Seljan and Dunđer [18]. More recently, Inaguma et al. [12] evaluated combining ASR and MT for multilingual speech translation on Ted Talks. Recent research has explored various methods for integrating ASR and MT to enable speech translation. Approaches include multitask learning to optimize the models jointly [1], passing attention information between encoder-decoders [22], combining CTC loss and attention [9], and knowledge distillation during training [14]. These techniques have achieved state-of-the-art results, showing promise for improving speech translation quality. However, performances still suffer on informal, conversational speech with terminology. Word error rate impacts MT quality more than speaker accents. Anastasopoulos and Chiang [1] proposed a tied multitask learning approach to optimize ASR and MT models jointly for speech translation.

Sperber et al. [22] introduced an attention-passing mechanism between the encoderdecoder of cascaded ASR and MT models for speech translation. Gaido et al. [9] combined CTC loss for the ASR model with attention based NMT for streaming speech translation. Liu et al. [14] proposed knowledge distillation during training to improve cascading of ASR and MT models for speech translation. This also achieved state-of the-art results.

3 Methodology

This research presents a comprehensive approach to creating a multilingual dataset in Arabic and Croatian from sports videos through an innovative data processing pipeline using automatic speech recognition and machine translation. The research is performed for two language pairs, i.e. English-Arabic and English-Croatian. The proposed data pipeline is based on post-editing and segmentation and is in fact a multi-phase processing approach that includes automatic caption generation, text corrections, and text segmentation.

Quality evaluation was performed on raw machine translations, then with post-editing, and afterwards with segmentation of text into logical units and sentence breaks. During this process, human assessments and BLEU scores were used to measure translation quality and the suitability of integrating ASR and MT in translating live sports commentary.

The experimental dataset consists of captions extracted from three videos related to football matches played during a football world cup event organized by FIFA. The selected videos were recorded during football matches in real time, in an ambient with background noise, expressions of excitement, interrupted speech, use of slang and domain-specific terminology.

The ASR system used was Google's commercial system. While the exact details of the ASR model architecture and training data are proprietary to Google, it likely utilizes an end-to-end deep neural network structure. State-of-the-art ASR systems like this are commonly trained on thousands of hours of transcribed multi-speaker conversational speech to learn acoustic and language models for converting speech to text [10, 16]. The models contain components to handle speaker variability as well as background noise robustness. However, challenges remain with informal speech containing disfluencies, interruptions, and unclear pronunciations.

The BLEU metric used in this paper calculates n-gram precision between the machine translations and reference translations, with a penalty for shorter length [17]. It is widely used for MT evaluation due to its speed and correlation with human judgments but has limitations like only capturing lexical similarity and favoring literal translation [4]. Captions in this research were generated by the Google Automatic Speech Recognition software for the English language, and then machine translated in three different phases, using the well-known online machine translation system Google Translate. Human and automatic evaluation is performed after each phase, and then correlated with respect to language pairs, iterations of phase improvement, and type of evaluation.

Machine translation throughout all three phases is performed for two language pairs, English-Arabic and English-Croatian.

3.1 Iterations and Phase Improvements

Machine translation is carried out throughout three different phases, where the next phase is based on the improvement of the previous one. The input for the machine translation system in the first phase (so-called Phase 1) derives from the process of generating English captions by using automatic speech recognition software. Subsequent phases encompassed improvements of text (so-called Phase 2), and improvements in segmentation (so-called Phase 3). More details are given below.

- Phase 1: automatic generation of captions in English by automatic speech recognition (ASR) software, combined with direct machine translation of captions without any text editing.
- Phase 2: editing of automatic captions of the first phase with improvements of text (lexical and syntactic corrections, fixing punctuation, capital letters etc.), combined with machine translation of post-edited captions.
- Phase 3: additional caption segmentation of the second phase (segmenting into logical units and sentence breaks), followed by machine translation of post-edited and segmented captions.

For the text corrections in Phase 2, examples include fixing misrecognized words (e.g. "lead" to "led"), adding punctuation, and correcting grammar errors. The segmentation

in Phase 3 separated caption text into logical utterances, with rules based on pauses, sentence structure, and semantic completeness.

3.2 Dataset

The dataset in this research consists of a total of 664 sentences or segments (text lines which do not end with a sentence delimiter), and all were machine translated:

- 102 sentences for Phase 1 (English-Arabic)
- 102 sentences for Phase 2 (English-Arabic)
- 128 sentences for Phase 3 (English-Arabic)
- 102 sentences for Phase 1 (English-Croatian)
- 102 sentences for Phase 2 (English-Croatian)
- 128 sentences for Phase 3 (English-Croatian)

Since the recorded speech in the videos included pauses, additional pre-processing and transcription was performed, which resulted in the rise of captions and, therefore, also in the rise of the number of sentences in Phase 3.

3.3 Evaluation

Evaluation is performed after each iteration (phase) by means of human quality evaluation and by using the automatic evaluation metric BLEU.

Human quality evaluation is performed by three native Arabic and three native Croatian speakers, for each language pair respectively. This type of evaluation is conducted with respect to the usability criterion and on a [1, 5] scale meaning:

- 5: Excellent (best grade, one small error is tolerated, such as a missing article, which essentially does not influence the meaning of a sentence/segment),
- 4: Very good (there are only a few errors, but which are not of critical importance),
- 3: Good (not very good, but not very bad either; there are more errors to correct, but machine translations are still understandable),
- 2: Enough (hardly enough to understand the context and meaning),
- 1: Not enough (worst grade, unintelligible machine translations).

The automatic evaluation is performed using the BLEU evaluation metric after each iteration. BLEU (BiLingual Evaluation Understudy) is an automatic metric, which is most frequently used for evaluating machine translation quality. BLEU matches machine translated n-grams with n-grams of its reference translation and counts the number of matches [17]. Most often it does it for 1–4 n-grams, which are all assigned the same weights. Then the geometric mean of the n-gram precision in a machine translation is calculated, based on a reference translation. A brevity penalty is also assigned to machine translations that are shorter than the corresponding reference translation.

In this research, BLEU scores are calculated and analyzed after each iteration and for each language pair and then correlated. For each iteration, a reference set had to be manually created first in order to calculate the BLEU scores.

4 Results and Discussion

Results of human quality evaluation show improvements from Phases 1 up to 3 for both language pairs, as presented in Fig. 1.

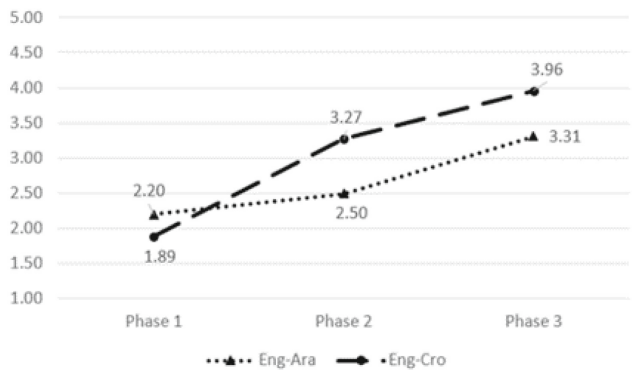

Fig. 1. Human evaluation – average scores of human quality evaluation of machine translated audio-video content for English-Arabic and English-Croatian.

Phase 1, which was obtained by means of automatic speech recognition of English and direct, i.e. raw machine translation into Arabic and Croatian performed by Google Translate, obtained the lowest scores, i.e. 2.2 for Arabic and 1.89 for Croatian.

Phase 2, in which the English original text was manually corrected and then machine translated, obtained higher scores – for Arabic scores they generally increased by 0.3 and for Croatian by ca. 1.4.

Phase 3, in which segmentation into logical units of the English text was performed, increased quality scores for both language pairs when compared to Phase 2 – for Arabic by 0.8 and for Croatian by 0.7.

Figure 2a presents human evaluation results in terms of average scores for three videos throughout the aforementioned three phases for English-Arabic. In the second phase, which included post-editing of Phase 1, the scores generally increased by 0.3 points, except in the first video. In Phase 3, which included logical segmentation, i.e. segmentation that follows the speed of speech, the scores increased by 0.6. When compared to Phase 1, the scores in Phase 3 generally increased by 1.0 point, hereby emphasizing the importance of correct speech recognition and logical segmentation.

For the English-Arabic language pair, errors were related to the specific football terminology (e.g. penalty [عقوبة], lead [الزعامة]), use of second meanings (e.g. gift horse [حصان هدية], good cross [تمريرة جيدة] etc.), proverbs (e.g. Don't look a gift horse in the mouth [لا تنظر إلى أسنان الحصان الهدية]), phrasal verbs (e.g. got a turn of pace [حصل على دور من السرعة]), verbs (e.g. wriggle [يتموج]), pronouns (e.g. himself [نفسه]), and recognition of proper names (e.g. Raketić [راكيتيتش]).

Figure 2b below presents human evaluation results in terms of average scores for three videos throughout the aforementioned three phases for English-Croatian, showing improvements from Phases 1 up to 3. Scores in Phase 2 improved on average by 1.2 points and there is again a substantial increase in Phase 3.

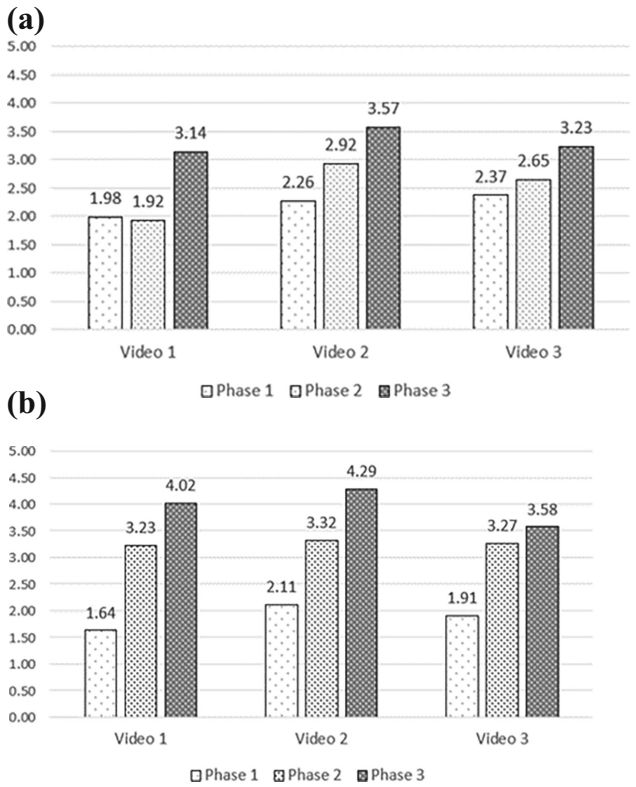

Fig. 2. (a) Human evaluation – average scores for three videos throughout the three phases for English-Arabic. (b) Human evaluation – average scores for three videos throughout the three phases for English-Croatian.

For the English-Croatian language pair errors were related to the specific football terminology (penalty, squad, level, cross, post etc.), use of second meanings (seesaw game, box, fashion, bedraggles etc.), phrases (get in, in the back etc.), proverbs, idioms (e.g. wriggle a little bit of space, just a fingertip away etc.), phrasal verbs (such as, pass up), use of pronouns (e.g. he, it) and recognition of proper names.

Figure 3a presents results of automatic quality evaluation using the popular BLEU metric throughout the three phases for English-Arabic, showing improvements through all three phases, and underlining the beneficial influence of text correction (improvement of Phase 2 with regard to Phase 1) and the influence of segmentation into logical units (in Phase 3 with regard to Phase 2). This improvement is evident in all three videos.

Figure 3b presents results of automatic quality evaluation using the BLEU metric throughout the three phases for English-Croatian. BLEU scores show considerable variations, although generally there are improvements throughout the phases.

The initial phase (Phase 1) achieved relatively low BLEU scores, pointing to problems of automatic speech recognition, where the ASR system did not recognize personal names, numbers, and proverbs. After applying corrections to the English text, BLEU results have improved, as shown in Phase 2. In Phase 3, results are slightly improved or diminished, still showing mostly benefits of logical segmentation.

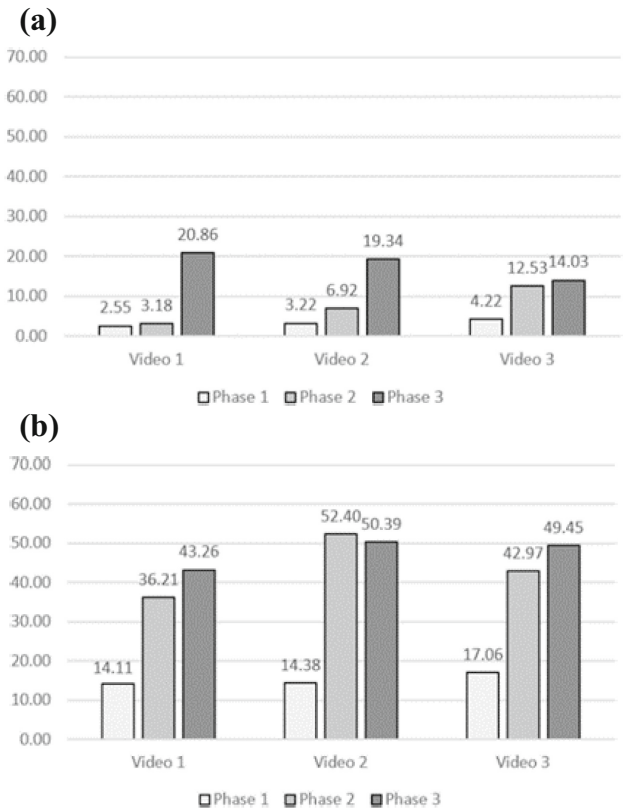

Fig. 3. (a) Automatic evaluation (BLEU) – average scores for three videos throughout the three phases for English-Arabic. (b) Automatic evaluation (BLEU) – average scores for three videos throughout the three phases for English-Croatian.

5 Limitations and Ideas for Future Work

This study has several limitations that provide opportunities for future work in order to provide comprehensive insights into the benefits and limitations of the proposed pipeline and the resulting multilingual corpus.

First, the dataset was relatively small, with only three videos. Additional experiments on larger and more diverse sports commentary data could further validate the findings. Second, only one generic ASR system and MT system were utilized. Testing custom models adapted to the sports domain could help overcome issues like terminology translation. Third, BLEU was sufficient for automatic evaluation in this study. However, the authors will consider using other automatic evaluation metrics such as METEOR, TER or PER to provide a more nuanced assessment of translation quality in each phase. In addition, only two language pairs were included. Expanding the language diversity could reveal new challenges.

These limitations emphasize the necessity for further experiments to fully assess the significance of the proposed pipeline and the resulting multilingual corpus. Future studies should also include comparative analyses of this corpus with existing state-of-the art corpora and evaluating its effectiveness for training and testing various machine translation models, as such comparisons and implementations are crucial for demonstrating the utility and efficacy of the proposed approach.

6 Conclusion

The aim of this paper was to evaluate the phases in a proposed data pipeline that combined automatic speech recognition and machine translation to translate audio-video content in the sports domain, specifically football matches.

In order to provide a deeper analysis of the real-world sports commentary dataset and to evaluate the robustness of the pipeline, evaluation was carried out by human judges and by applying the automatic BLEU metric to translations of three videos, throughout three iterative phases of processing, for the language pairs English-Arabic and English-Croatian.

In Phase 1, raw English output from an automatic speech recognition system was machine translated into the target languages without any editing. For Phase 2, the ASR transcript was manually corrected by human experts before machine translation. Finally in Phase 3, the corrected English text was further segmented into logical utterance units following the original speech patterns.

The results revealed several key challenges and complexities in applying general purpose ASR and MT technologies to translate live sports commentary speech. ASR issues arising from background noise, oral discourse features like interruptions and hesitations, and unrecognized domain terminology, numbers and proper names led to lower quality scores in Phase 1. After substantial text correction and formatting, translation quality significantly improved for both target languages in Phase 2. However, some problems persisted relating to accurate translation of sports terms, idioms, proverbs and slang expressions. On average, human evaluation scores increased by 1 point on the 5point scale between Phase 1 and Phase 3 for English-Arabic, and by 0.91 points for English-Croatian. This quantitatively demonstrates the benefits of proper speech recognition and logical segmentation of the source text before translation.

The trends in automated BLEU metric scores aligned closely with the human judgments, with absolute gains of +0.31 for English-Arabic and + 0.11 for English-Croatian resulting from the process enhancements in Phases 2 and 3. The metrics corroborated that issues remained even after processing, including terminology translation errors and challenges parsing long, complex phrases.

Human evaluation correlated strongly between language pairs (0.80) but showed some disagreement likely due to structural differences. Human scores also correlated well with BLEU, at 0.88 for English-Arabic and 0.91 for English-Croatian. BLEU reflected quality improvements across phases, with gains of +0.31 for Arabic and +0.11 for Croatian. This quantitatively demonstrates the cumulative benefit of corrections and segmentation. However, issues remained like terminology errors and complex phrases. In summary, the experimental results confirmed significant difficulties in directly applying generic, general-purpose ASR and MT systems to accurately translate informal, interrupted live sports commentary speech containing domain-specific vocabulary. While the multi-phase post-processing corrections and segmentation of the raw ASR output did improve downstream machine translation output, there were lingering issues around recognition and translation of key terms, names and complex expressions. Limitations of the current study include the relatively small dataset size and lack of model customization. To address these limitations, future work should investigate expanding the

language pairs, audio-video content, and exploring tailoring the ASR and MT models to the sports commentary domain.

In addition, more evaluators and automatic metrics can be applied as a way of increasing the significance of research results. The authors plan to investigate the applicability of the proposed multi-phase data processing pipeline to other language pairs that have different syntactic language structures in order to verify the potential of enhancing translation quality over successive phases.

Overall, the findings emphasize that ASR quality greatly impacts subsequent machine translation, as noise, unfinished sentences and unclear pronunciations pose challenges for the pipeline. Even though the MT system performed relatively well on its own, it was insufficient to enable high-fidelity information transfer between the significantly different language families of English, Arabic and Croatian. Beyond structural differences, the abundant domain terminology differences created further difficulties.

Nevertheless, since substantial improvements were observed between successive phases, the authors plan to increase the volume of the parallel corpus (dataset) developed in this work, and to compare customized MT models to the general system evaluated here, to address the challenging problem of translating live sports commentary, by incorporating and improving the integrated ASR-MT pipeline.

7 Ethics Statement

This research was conducted with strong consideration for ethical practices. Participation by human annotators and evaluators was completely voluntary. All participants were provided with details on the purpose of the study and how their contributions would be utilized. Formal consent was obtained from participants prior to collecting any human judgments. The privacy and confidentiality of the participants was ensured by making their evaluation responses anonymous. No personally identifiable information was collected from evaluators involved in annotation, post-editing or quality rating tasks. The video data used was publicly available sports commentary content.

In reporting experimental results, care has been taken not to make unsupported claims beyond what the data warrants. The limitations of the methodology and findings are clearly acknowledged. Credit for tools and resources used has been properly attributed. The multilingual dataset compiled and generated for this study will be shared publicly to spur further research and development of speech translation technologies. By providing open access to the data, the authors aim to make the benefits of this work available to all.

Acknowledgments. This publication was partially made possible by grant number NPRP14C-0916-210015 from the Qatar National Research Fund/Qatar Research Development and Innovation Council (QRDI). The contents herein reflect the work and are solely the authors' responsibility.

Data Availability. The multilingual dataset generated in this study will be made publicly available upon publication, contributing new resources for research in sports commentary speech translation. The videos and the dataset contain the English speech transcripts produced by the ASR

system, along with the machine translations and human reference translations in Arabic and Croatian. By releasing the data, the authors aim to promote further advancement and benchmarking of ASR and MT methods on this challenging application.

References

1. Anastasopoulos, A., Chiang, D.: Tied multitask learning for neural speech translation. In: Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL), vol. 1, pp. 82–91. Association for Computational Linguistics (ACL), New Orleans (2018)
2. Armstrong, S., Caffrey, S., Flanagan, M.: Translating DVD subtitles from English-German and English-Japanese using example-based machine translation. In: Proceedings of the MuTra 2006 Conference – Audiovisual Translation Scenarios, pp. 1–13. Advanced Translation Research Center (ATRC), Copenhagen (2006)
3. Burchardt, A., Lommel, A., Bywood, L., Harris, K., Popovic, M.: Machine translation quality in an audiovisual context. Target **28**(2), 206–221 (2016)
4. Callison-Burch, C., Osborne, M., Koehn, P.: Re-evaluating the role of bleu in machine translation research. In: Proceedings of the 11th Conference of the European Chapter of the Association for Computational Linguistics, pp. 249–256. Association for Computational Linguistics (ACL), Trento (2006)
5. Dunđer, I.: Machine translation system for the industry domain and Croatian language. J. Inf. Organiz. Sci. **44**(1), 33–50 (2020)
6. Dunđer, I., Seljan, S., Pavlovski, M.: Automatic machine translation of poetry and a LowResource language pair. In: Proceedings of the 43rd International Convention on Information, Communication and Electronic Technology (MIPRO2020), pp. 1034–1039. Croatian Society for Information, Communication and Electronic Technology (MIPRO), Opatija (2020)
7. Dunđer, I., Seljan, S., Pavlovski, M.: What makes machine-translated poetry look bad? A human error classification analysis. In: Proceedings of the Central European Conference on Information and Intelligent Systems (CECIIS 2021), pp. 183–191. Faculty of Organization and Informatics, Varaždin Online (2021)
8. Etchegoyhen, T., et al.: Machine translation for subtitling: a large-scale evaluation. In: Proceedings of the Language Resources and Evaluation (LREC 2014), pp. 46–53. European Language Resources Association (ELRA), Reykjavik (2014)
9. Gaido, M., Di Gangi, M.A., Negri, M., Turchi, M.: End-to-end speech-translation with knowledge distillation: FBK@IWSLT2020. In: Proceedings of the 17th International Conference on Spoken Language Translation, pp. 80–88. Association for Computational Linguistics (ACL), online (2020)
10. Gales, M., Young, S.: The application of hidden Markov models in speech recognition. Found. Trends Signal Process. **1**(3), 195–304 (2007)
11. Habash, N.Y.: Introduction to Arabic Natural Language Processing. Synthesis Lectures on Human Language Technologies. Springer, Cham (2010)
12. Inaguma, H., Duh, K., Kawahara, T., Watanabe, S.: Multilingual end-to-end speech translation. In: Proceedings of the 2019 IEEE Automatic Speech Recognition and Understanding Workshop (ASRU), pp. 570–577. IEEE, Sentosa (2019)
13. Koehn, P.: Statistical Machine Translation. Cambridge University Press, Cambridge (2009)
14. Liu, Y., et al.: End-to-end speech translation with knowledge distillation. In: Proceedings of the Conference Interspeech 2019, pp. 1128–1132. International Speech Communication Association (ISCA), Graz (2019)
15. Martín-Mor, A., Sánchez-Gijón, A.: Machine translation and audiovisual products: a case study. J. Specialised Transl. **26**, 172–186 (2016)

16. McGraw, I., et al.: Personalized speech recognition on mobile devices. In: Proceedings of the 2016 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), pp. 5955–5959. IEEE, Shanghai (2016)
17. Papineni, K., Roukos, S., Ward, T., Zhu, W.-J.: BLEU: a method for automatic evaluation of machine translation. In: Proceedings of the 40th Annual Meeting of the Association for Computational Linguistics (ACL 2002), pp. 311–318. Association for Computational Linguistics (ACL), Philadelphia (2002)
18. Seljan, S., Dunđer, I.: Combined automatic speech recognition and machine translation in business correspondence domain for English-Croatian. J. Comput. Inf. Syst. Control Eng. **8**, 1069–1075 (2014)
19. Seljan, S., Dunđer, I.: Automatic quality evaluation of machine-translated output in sociological-philosophical-spiritual domain. In: Proceedings of the 10th Iberian Conference on Information Systems and Technologies (CISTI 2015), vol. 2, no. 1, pp. 128–131. CISTI, Aveiro (2015)
20. Seljan, S., Katalinić, J.: Integrating localization into a video game. In: Proceedings of the conference INFuture 2017: Integrating ICT in Society, pp. 43–55. Department of Information and Communication Sciences, Faculty of Humanities and Social Sciences, University of Zagreb, Zagreb (2017)
21. Seljan, S., Dunđer, I., Pavlovski, M.: Human quality evaluation of machine-translated poetry. In: Proceedings of the 43rd International Convention on Information, Communication and Electronic Technology (MIPRO2020), pp. 1040–1045. Croatian Society for Information, Communication and Electronic Technology (MIPRO), Opatija (2020)
22. Sperber, M., Neubig, G., Niehues, J., Waibel, A.: Attention-passing models for robust and data-efficient end-to-end speech translation. Trans. Assoc. Comput. Linguist. **7**, 313–325 (2019)
23. Volk, M.: The automatic translation of film subtitles. A machine translation success story? J. Lang. Technol. Comput. Linguist. **24**(3), 115–128 (2009)
24. Kraif, O., Laurent, D., Nguyen, T.M.H., Semmar, N., Stuck, F., Zaghouani, W.: This paper might provide foundational insights into evaluation campaigns related to machine translation and could offer a historical perspective or methodological insights beneficial for your study (2005)
25. Alam, F., et al.: The work on check-worthiness in multimodal content can provide insights into integrating various language technologies, which might be relevant to your study's focus on combining speech recognition and translation (2023)
26. Atwell, E., et al.: This paper discusses understanding the Quran through computing methods, which, while not directly related, might provide insights into complex language processing tasks that could inform your methodology, especially for Arabic (2010)
27. Draffan, E., et al.: While focused on AAC users, this paper could offer insights into speech technology applications that may enrich your research's discussion on speech recognition (2015)
28. Schlippe, T., Alessai, S., El-Taweel, G., Wölfel, M., Zaghouani, W.: This research on visualizing voice characteristics might provide a unique perspective on processing speech data, which could be relevant for your study's automatic speech recognition component (2020)
29. Seljan, S., Zaghouani, W.: Discusses integrating machine translation into a webbased environment, which seems directly relevant to your study's focus on a web-based audio-video environment (2019)
30. Zaghouani, W., Habash, N., Obeid, O., Mohit, B., Bouamor, H., Oflazer, K.: Building an Arabic machine translation post-edited corpus: guidelines and annotation. In Proceedings of the Tenth International Conference on Language Resources and Evaluation (LREC 2016), pp. 1869–1876. European Language Resources Association (ELRA), Portorož (2016)

31. Temnikova, I., Wajdi, Z., Vogel, S., Habash, N.: Applying the Cognitive Machine Translation Evaluation Approach to Arabic (2016)
32. Zaghouani, W., Temnikova, I.: Toward a cognitive evaluation approach for machine translation PostEditing. Qatar Found. Annu. Res. Conf. Proc. **2018**(3), ICTPD885 (2018)
33. Zaghouani, W., Abdelali, A., Guzman, F., Sajjad, H.: Normalizing mathematical expressions to improve the translation of educational content. In: Proceedings of the AMTA 2016 Workshop Semitic Machine Translation (SeMaT) Collocated with EMNLP 2016 Workshops on November 1st (2016)